

Brain Tumor Classification from MRI: A Comparative Study of Convolutional Neural Networks and Vision Transformers

Enci Zhang, Lufei Wang, Tony Hou, Jessica Nickell

Problem Description

Brain tumors are an abnormal growth of cells in the brain that can be indicative of several severe underlying illnesses and have an array of serious symptoms such as headaches, vision issues, speech issues, etc. Magnetic Resonance Imaging (MRI) is a significant part of the diagnostic process. It produces 3D images of the brain and can help identify and diagnose brain tumors. The early identification of these brain tumors can have a direct impact not only on the patient's estimated survival but also on treatment planning and tumor growth evaluation. Automated classification of brain MRI images is a challenging image recognition task that requires classification between categories with similar visual characteristics. Variations in tumor appearance and image quality can make accurate classification challenging. In terms of predictive accuracy, computational requirements are an important consideration when selecting an appropriate deep learning model. These factors motivate a systematic comparison of architectures with different approaches to how they extract and combine image features. This project addresses the classification of two-dimensional brain MRI images into four categories: glioma, meningioma, pituitary tumor, and no tumor, using the Brain Tumor MRI Dataset containing 7,200 images.

Related Work

Convolutional neural networks (CNNs) are widely used for image classification due to their capacity to learn hierarchical visual representations. Recent studies have shown that deep learning has achieved strong performance in brain tumor classification using MRI images. Mzoughi et al. (2020) proposed a multi-scale 3D convolutional neural network that uses volumetric MRI data to classify glioma tumors. Their work demonstrated that 3D CNNs can effectively capture spatial information across MRI slices, but they also require more computational resources.

He et al. (2016) introduced residual learning, which facilitates the optimization of deep networks through shortcut connections. This framework provides the foundation for ResNet-50, selected in the present study as an established CNN baseline. Tan and Le (2019) proposed EfficientNet, a family of architectures developed by balancing network depth, width, and input resolution. EfficientNet-B0, one model of the family, is introduced to examine the trade-off between classification performance and computational cost within a compact convolutional architecture.

Transformer architectures offer another approach to image classification by using attention to model relationships between image patches. Touvron et al. (2021) introduced Data-Efficient Image Transformers (DeiT), which shows competitive image classification performance through training on ImageNet without additional external training data. Pretrained DeiT-Small is included to investigate performance of a vision Transformer as an alternative to convolutional feature extraction for brain MRI classification tasks.

Liu et al. (2021) proposed Swin Transformer, which constructs hierarchical representations using self-attention within local windows. The shifted-window mechanism enables information exchange across window boundaries, while the hierarchical architecture supports feature representation at multiple spatial scales. Swin-Tiny is included to examine whether this attention structure offers advantages over the global attention employed by DeiT-Small for the selected classification task.

Proposed Work

Inspired by these architectural developments, our project aims to compare pretrained ResNet-50, EfficientNet-B0, DeiT-Small, and Swin-Tiny in classification performance and computational efficiency on the brain tumor dataset. We will explore the strengths and limitations of models and identify trade-offs between predictive performance and computational cost. Although these foundational studies demonstrate the models' capabilities on general image recognition tasks, their relative performance on the MRI dataset remains unknown. In our experiment, ResNet-50 and EfficientNet-B0 represent convolutional architectures, while DeiT-Small and Swin-Tiny each represent global and hierarchical attention-based architectures. All models will be initialized using ImageNet-1K-pretrained weights and fine-tuned on the selected MRI dataset. The standard, non-distilled DeiT-Small configuration will be used. The original classification layer of each pretrained model will be replaced with a new layer containing 4 output units, corresponding to glioma, meningioma, pituitary tumor, and no tumor.

Images will be resized to 224×224 pixels and represented using three input channels. Normalization will follow the requirements of each pretrained model. An augmentation policy of rotations, translations, and zoom transformations will be applied to training images. We will use the dataset's existing training and test sets as the initial split, reserving 20% of the training images for validation while preserving class proportions. Duplicate images across sets will be checked before training. All models will use identical data splits, the same number of hyperparameter trials, and three fixed random seeds.

Evaluation Metric

Model performance will be evaluated on the held-out test set after hyperparameters and training checkpoints have been selected using the validation set. Macro-averaged F1 score (macro-F1) will serve as the primary evaluation metric since it combines precision and recall while assigning equal importance to all four classes. It is defined as

$$\text{Macro-F1} = \frac{1}{4} \sum_{c=1}^4 \frac{2TP_c}{2TP_c + FP_c + FN_c},$$

where TP_c , FP_c , and FN_c denote the numbers of true positives, false positives, and false negatives for class c , respectively.

Overall accuracy will measure the proportion of correctly classified images. Class-specific precision and recall will provide further insight into each model's performance. Specifically, precision measures the proportion of predictions for a class that are correct, and recall measures the proportion of images belonging to that class that are correctly identified. A confusion matrix normalized by true class will illustrate misclassification patterns across the four categories.

Each model will be trained using three random seeds and identical data partitions. Classification metrics will be reported as the mean and standard deviation across these runs to show variation due to training randomness.

Computational efficiency will be evaluated using the total number of model parameters, average training time per epoch, and average inference time per image. These metrics measure model size, training speed, and prediction speed respectively. Runtime measurements will use the same hardware, input resolution, and numerical precision, with a common training batch size and an inference batch size of one.

References

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